

A266 – Unit Check and SI Back-Computation

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.1 Purpose

[STIPULATION] In FFGFT all formulas are set up in natural units ($c = \hbar = k_B = 1$). When back-computing to SI quantities, the physical constants reappear with specific powers. These powers are *not* freely choosable – they follow necessarily from dimensional analysis. An error in the power of c is an error of factors $\sim 3 \times 10^8$ to $(3 \times 10^8)^4$ – not noticeable when looking only at orders of magnitude, fatal when comparing with a measurement.

This document gives the systematic procedure: first dimensional analysis in natural units, then automated back-computation, then verification. It is not a teaching document on unit systems (that is A080/A085) – it is a practical toolbox for concrete calculations.

.2 Basic Principle: All Dimensions as Energy Powers

[B] In natural units $c = \hbar = k_B = 1$. Every physical quantity thereby has a unique dimension $[E]^n$ with rational n :

Quantity	Nat. dim.	SI origin	n
Energy	$[E]^1$	—	+1
Mass	$[E]^1$	$E = mc^2 \Rightarrow m = E/c^2$	+1
Momentum	$[E]^1$	$p = E/c$	+1
Length	$[E]^{-1}$	$\ell = \hbar c/E$	-1
Time	$[E]^{-1}$	$t = \hbar/E$	-1
Velocity	$[E]^0$	$v = \ell/t = c = 1$	0
Force	$[E]^2$	$F = E/\ell = E^2/(\hbar c)$	+2
Pressure/energy density	$[E]^4$	$\rho = E/\ell^3 = E^4/(\hbar c)^3$	+4
Charge	$[E]^0$	from $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ dimensionless	0
Temperature	$[E]^1$	$T = E/k_B$	+1
Entropy	$[E]^0$	$S = k_B \cdot (\text{dimensionless})$	0
Gravitational constant G	$[E]^{-2}$	$G = \ell_p^2 c^3/\hbar$	-2

[B] Rule. From n the back-computation factor follows directly: a quantity with $[E]^n$ in natural units has the SI value

$$X_{\text{SI}} = X_{\text{nat}} \cdot \left(\frac{\hbar c}{1 \text{ SI-length}} \right)^n.$$

In practice: assemble the SI factor from $\hbar$, c , k_B so that the SI dimension comes out.

.3 The c -Power in Back-Computation: Systematic Derivation

[B] The question “which power of c ?” is answered from the SI dimension of the target quantity. The method in three steps:

Step 1: Write down the SI target dimension. E.g. energy density: $[\rho] = \text{J/m}^3 = \text{kg m}^{-1} \text{s}^{-2}$.

Step 2: Express SI dimension through $\hbar^a c^b k_B^d$. $\hbar$ has SI dim $[\text{J s}]$, c has $[\text{m/s}]$, k_B has $[\text{J/K}]$. Find $a, b, d \in \mathbb{Z}$ such that $[\hbar^a c^b k_B^d] = \text{target dim}/[E_{\text{nat}}^n]$.

Step 3: Verification by substitution. Insert the found (a, b, d) and check against the measured SI quantity.

[K] Table of most common back-computation factors:

Quantity	Nat. unit	SI factor	Derivation
Mass	$[E]$ in MeV	$\div c^2$	$m_e = 0.511 \text{ MeV}/c^2$
Length	$[E]^{-1}$	$\times \hbar c$	$\ell_p = 1/E_p \cdot \hbar c$
Time	$[E]^{-1}$	$\times \hbar$	$t_p = 1/E_p \cdot \hbar$
Velocity	dimensionless	$\times c$	$v = 0.5 \cdot c$
Force	$[E]^2$	$\div (\hbar c)$	$F[\text{N}] = F_{\text{nat}} \cdot E^2/(\hbar c)$
Energy density	$[E]^4$	$\div (\hbar c)^3$	$\rho[\text{J/m}^3] = \rho_{\text{nat}} \cdot E^4/(\hbar c)^3$
Temperature	$[E]$	$\div k_B$	$T[\text{K}] = E[\text{J}]/k_B$
Frequency f	$[E]^1$	$\div \hbar$	$f[\text{s}^{-1}] = E[\text{J}]/\hbar$
G	$[E]^{-2}$	$\times \hbar c^5$	$G[\text{m}^3/\text{kg s}^2] = G_{\text{nat}}[\text{J}^{-2}] \cdot \hbar c^5$

.4 Concrete Examples from FFGFT

Mass: One Power of c^2

[K] From $E = mc^2$ it follows that $m = E/c^2$. A value in natural units (MeV) becomes a SI mass by dividing by c^2 :

$$m_e = 0.511 \frac{\text{MeV}}{c^2} = \frac{0.511 \times 10^6 \times 1.602 \times 10^{-19} \text{ J}}{(2.998 \times 10^8 \text{ m/s})^2} = 9.109 \times 10^{-31} \text{ kg}.$$

Error c^1 instead of c^2 : factor 3×10^8 off. Error c^0 (no c): factor $(3 \times 10^8)^2 \approx 10^{17}$ off.

Planck Length: $\hbar c$ Together, Not c Alone

[K] ℓ_P in natural units is an inverse energy. Back-computation to metres needs $\hbar c$:

$$\ell_P = \frac{\hbar c}{E_P} = \frac{(1.055 \times 10^{-34} \text{ J s})(2.998 \times 10^8 \text{ m/s})}{1.956 \times 10^9 \text{ J}} = 1.616 \times 10^{-35} \text{ m}.$$

Only $\hbar/E_P$ (without c) gives the Planck *time*, not the Planck *length* – a typical error.

Energy to Inverse Time: Only $\hbar$, No c

[K] An energy E (dimension $[E]^1$) is back-computed to inverse time $[s^{-1}]$ by dividing by $\hbar$:

$$f = \frac{E}{\hbar}, \quad [f] = \frac{[\text{J}]}{[\text{J s}]} = [\text{s}^{-1}]. \checkmark$$

Example: $E = 1.41 \times 10^{-33} \text{ eV} = 2.26 \times 10^{-52} \text{ J}$ gives $f = E/\hbar = 2.14 \times 10^{-18} \text{ s}^{-1}$ (A265). Dividing by $\hbar c$ instead gives an inverse length (wrong dimension). The c -power is zero – it does not appear.

Energy Density: Four Powers of c in the Denominator

[K] Casimir energy density ρ has dimension $[E]^4$ in natural units. Back-computation:

$$\rho[\text{J/m}^3] = \rho_{\text{nat}} [E^4] \cdot \frac{1}{(\hbar c)^3}.$$

Three powers of $\hbar c$ in the denominator – because length contributes $[E]^{-1}$ three times. c appears as c^3 (from the three $\hbar c$).

Gravitational Constant: c^5 in the Numerator

[K] In natural units G has dimension $[E]^{-2}$. The SI constant $G = 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$ requires:

$$G_{\text{SI}} = G_{\text{nat}} \cdot \hbar c^5, \quad [G_{\text{SI}}] = [E]^{-2} \cdot [\hbar][c]^5 = \frac{\text{J s} \cdot \text{m}^5/\text{s}^5}{\text{J}^2} = \frac{\text{m}^5}{\text{J s}^4} = \frac{\text{m}^3}{\text{kg s}^2}. \checkmark$$

Check: $G_{\text{nat}} = 1/E_P^2$ with $E_P = 1.956 \times 10^9 \text{ J}$ gives $\hbar c^5/E_P^2 = 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$ (equivalent to $G = \hbar c/M_P^2$ with $M_P = E_P/c^2$). Error: c^3 instead of c^5 gives units of m/kg and a value too small by $c^2 \approx 9 \times 10^{16}$ – wrong dimension immediately recognisable.

.5 The General Procedure: Dimension Matrix

[B] For every formula $X = f(\xi, E_0, \dots)$ in natural units (here $X = f(\xi, E_0, \dots)$ is a general formula, not a frequency and not the base winding number $f = 1/\xi$; elsewhere in this document f stands for a frequency; cf. Doc. 190, R147):

1. Determine n from the table above (or by counting lengths/masses/times in the SI dimension of X).

2. Write the SI factor:

$$X_{\text{SI}} = X_{\text{nat}} \cdot \hbar^a \cdot c^b \cdot k_B^d$$

with a, b, d from the linear system:

$$[\text{kg}] : a + n = n_{\text{kg}}$$

$$[\text{m}] : 2a + b + 2n = n_{\text{m}}$$

$$[\text{s}] : -a - b - 2n = n_{\text{s}}$$

(where $n_{\text{kg}}, n_{\text{m}}, n_{\text{s}}$ are the exponents of the SI dimension of X ; k_B is added when kelvin appears.)

3. Verification: insert $\hbar = 1.055 \times 10^{-34} \text{ J s}$, $c = 2.998 \times 10^8 \text{ m/s}$, $k_B = 1.381 \times 10^{-23} \text{ J/K}$ and check the SI unit.

[K] When k_B appears and when not. k_B appears exactly when the target quantity contains kelvin (temperature, entropy/kelvin) or when converting from an energy to a kelvin quantity. In all other cases (masses, lengths, times, forces) k_B does not appear.

.6 Unit Check of a Formula

[B] Every formula set up in FFGFT should pass the unit check: both sides of the equation must have the same energy power $[E]^n$ in natural units, and on back-computation the same SI unit must emerge.

[K] Example: $T \cdot E = 1$ (A060). In natural units: $[T] = [E]^{-1}$ (time) and $[E] = [E]^{+1}$. Product: $[E]^0$ – dimensionless. The product $T \cdot E = 1$ is a correct equation between a time and an inverse time – the unit check is passed.

[K] Example: $\alpha = \xi \cdot E_0^2$ – **unit check of the α formula.** The formula is *correct*. The unit check shows why – and simultaneously makes the implicit reference scale visible.

α is dimensionless, ξ is dimensionless. The equation therefore holds only if E_0^2 is also dimensionless. That is the case when E_0 appears as a pure numerical ratio – i.e. measured in MeV and divided by 1 MeV:

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2.$$

The 1 MeV in the denominator is the *implicit reference scale* – it makes $E_0/1 \text{ MeV}$ dimensionless and the equation correct. It does not stand explicitly in the short notation, but is indispensable.

Numerical verification:

$$\alpha = \frac{4}{30000} \cdot \left(\frac{7.398 \text{ MeV}}{1 \text{ MeV}} \right)^2 = 1.333 \times 10^{-4} \times 54.73 = 7.297 \times 10^{-3} = \frac{1}{137.0}.$$

Agreement with experiment to 0.0005 % – the formula is correct.

What the unit check reveals: the choice of MeV as reference scale is not free – it is the empirical SI anchor of the FFGFT sector. A different reference scale (e.g. GeV) would give ξ a different numerical value, but α remains the same physical number. This is the

statement from A261: $E_0 = 7.398 \text{ MeV}$ carries the anchor, and therefore one may not simply insert different energy units in this formula without simultaneously changing ξ .

[S] Converting to natural units. Translating a SI formula to natural units: replace all $\hbar$, c , k_B by 1 and drop the units. The resulting formula is then valid only in natural units; on back-computation all factors return.

.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$[X]$	Dimension of quantity X
$[E]^n$	Energy dimension to power n
$n_{\text{kg}}, n_{\text{m}}, n_{\text{s}}$	Exponents of the SI unit of X in $\text{kg}^{n_{\text{kg}}} \text{m}^{n_{\text{m}}} \text{s}^{n_{\text{s}}}$
a, b, d	Exponents of $\hbar^a c^b k_B^d$ in the back-computation factor
$E_p = \sqrt{\hbar c^5 / G} \approx 1.956 \times 10^9 \text{ J}$	Planck energy

.8 Verification Script

- Numerical verification of all back-computation examples; dimension matrix solver for arbitrary SI target dimensions; verification for m_e , ℓ_P , inverse time, ρ_{Casimir} , G :
python/A_Serie_Skripte/a266_einheitenpruefung.py

.9 Open Questions

The dimension matrix presupposes that the formula is set up in pure Planck units. Mixed systems (e.g. MeV for energy, fm for length) require additional conversion steps.

Non-integer energy powers. Some FFGFT expressions have rational n (with $\hbar = c = 1$ charge is dimensionless, $[E]^0$, as the table above states from $\alpha = e^2 / (4\pi\epsilon_0 \hbar c)$). The procedure applies in principle, but the linear system then gives rational solutions.

.10 Supersedes

This document subsumes: Doc. 013 (dimensional analysis, gravitational constant from ξ , Planck length, conversion factors), Doc. 014 (natural units, dimension table, scaling factor S_{T0} , derivation vs. back-computation), Doc. 015 (unit conversions, energy dimension table, force/pressure/charge). *Not taken over*: pedagogical derivation of the individual constants from ξ (A015, A145) and the unit convention as such (A080, A085).

.11 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.